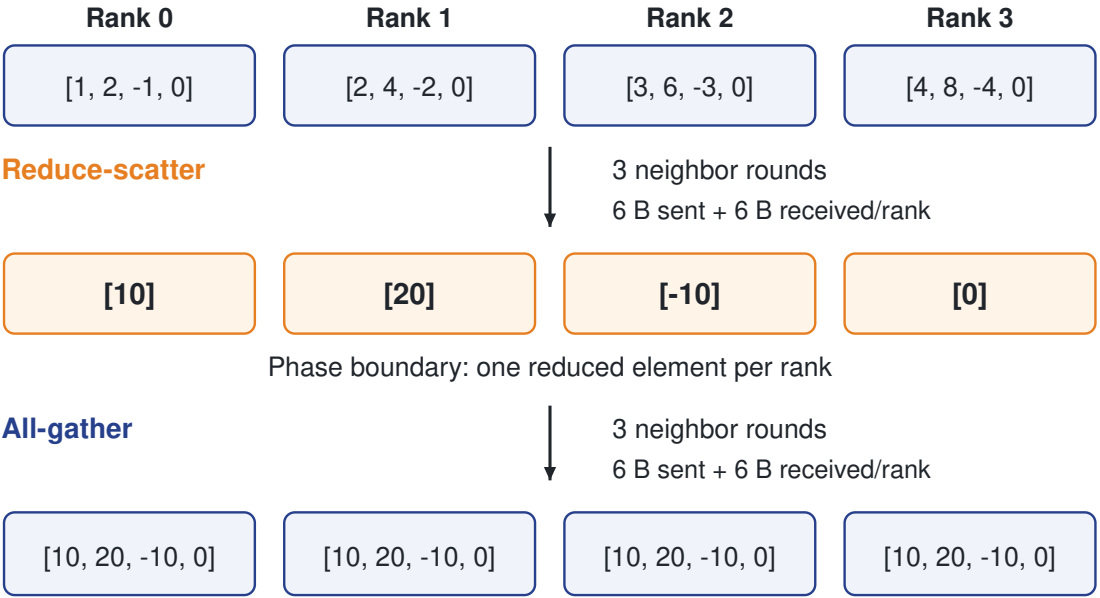
SUM first; replication second

Four ranks; four FP16 elements per input (8 B/rank).



Every rank holds SUM, not the average.

Balanced ring neighbor order (both phases)

Each round sends one 2 B chunk to the next rank.

Per rank: 12 B sent + 12 B received = 24 B endpoint total